DO NOW — RECALL

GPUs, multicore and parallel systems

Name: _____ Class: _____ Date: _____

1. Why doesn't a smaller assembly instruction count guarantee a faster program?

2. What is a core?

3. Two independent CPU-bound tasks each take 4 minutes when run on its own core. They start together on two separate identical cores. Calculate the ideal elapsed completion time, ignoring overhead.

4. Given int[] a={1,2,3,4}, perform a[i]=a[i]*2 for even indexes 0 and 2 only. Give final array and its sum.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Different instructions have different costs, and decoding, caches and pipelining affect performance.

2. Model answer

An execution unit capable of carrying out its own instruction stream.

3. Model answer

4 minutes. Both cores execute at the same time and each finishes its task after 4 minutes.

4. Model answer

Final [2,2,6,4], sum=14. Indexes 0 and 2 were changed, not the even-valued elements.

Vocabulary — know these terms

GPU — Processor designed for highly parallel workloads.

multicore — CPU containing multiple execution cores.

data parallelism — Apply the same operation to many independent data elements.

bandwidth — Volume of data transferred in a given time.

synchronisation — Coordinating shared work and results.

NEW LEARNING

Read, annotate and explain

GPUs and data parallelism

A GPU contains many execution resources suited to repeating similar calculations across graphics pixels or machine-learning tensors.

Multicore CPUs

Multiple CPU cores can run independent threads/processes, enabling greater throughput when tasks can be parallelised.

Evaluate overhead

Data transfers, branching, synchronisation and sequential tasks can reduce the ideal speedup. A GPU is not automatically best for every problem.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Compare 12 independent image tiles split across four cores under an ideal equal-work model.

Predict first

Divide 12 independent equal-sized tiles among four available cores.

Explain and check

Each core processes three tiles, so ideal completion is around one quarter of serial time only if work and overhead allow.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Give two uses of GPUs, including one not based on graphics, and explain a common characteristic of both.

Question 2

A task has 24 independent identical chunks and six CPU cores. Calculate ideal work per core and describe two constraints on real speedup.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A video-editing workflow must first decode frames sequentially, then apply the same filter independently to each frame, before assembling output. Explain which parts suit GPU execution and why.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

GPUs always execute any sequential algorithm faster than a CPU.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.